Comprehensive Report Analysis

Customer satisfaction and Segmentation reports based on the images generated :-

1. Class Distribution of Satisfaction Level

This bar graph shows the count of samples for each customer satisfaction level:

- **Satisfaction Level 1:** Approximately 117 samples.
- **Satisfaction Level 2:** Approximately 108 samples.
- **Satisfaction Level 3:** Approximately 125 samples.

Analysis: The distribution of satisfaction levels is fairly balanced. There isn't a severe class imbalance, which is beneficial for training unbiased classification models.

2. Correlation Heatmap of Features

This heatmap illustrates the Pearson correlation coefficients between various features and `Satisfaction Score`.

- **Strong Positive Correlations with `Satisfaction Score`:**
  - `Average Rating`: 0.94
  - `Total Spend`: 0.69
  - `Items Purchased`: 0.69
  - `Frequency`: 0.69
  - `Monetary`: 0.69
  - `Tenure`: 0.61
  - `Membership Type_Gold`: 0.70
- **Strong Negative Correlations with `Satisfaction Score`:**
  - `Age`: -0.58
  - `Membership Type_Silver`: -0.50
- **High Inter-Feature Correlations:** Features like `Total Spend`, `Items Purchased`, `Frequency`, and `Monetary` are highly correlated with each other (values near 1.00), suggesting they capture similar aspects of purchasing behavior. `Average Rating` also shows high correlations with these spending-related features and `Tenure`.

Analysis: These strong correlations indicate that features related to spending habits, average ratings, and customer tenure are highly predictive of customer satisfaction. `Age` also plays a significant role, but negatively.

3. Age vs Total Spend

This scatter plot visualizes the relationship between `Age` and `Total Spend`.

- The plot shows distinct clusters of data points rather than a continuous linear relationship. For instance, there are concentrated groups of customers at certain age and spending ranges.

Analysis: The clustered nature suggests that customer segments might exist based on age and spending patterns, which is a good precursor for clustering.

4. Age vs Satisfaction Level

This box plot displays the distribution of `Age` across different `Satisfaction Levels`.

- **Satisfaction Level 0:** Median age around 36.
- **Satisfaction Level 1:** Median age is lower, around 30, with a tighter spread. ● **Satisfaction Level 2:** Median age around 36.5, but with a much broader age range.

Analysis: This highlights that customers with `Satisfaction Level 1` tend to be younger on average, supporting the negative correlation between `Age` and `Satisfaction Score` seen in the heatmap.

5. Customer Segments (PCA 2D Projection)

This scatter plot projects customer data onto two principal components and colors them by their assigned clusters.

- The plot clearly shows **four distinct and well-separated clusters** (labeled 0, 1, 2, and 3).
- The separation indicates that the clustering algorithm has effectively grouped similar customers together.

Analysis: The visual separation of the clusters supports a good clustering result, as also indicated by your previously reported Silhouette Score of 0.593.

6. Customer Segment Characteristics

These two bar charts illustrate the average `Total Spend` and `Average Satisfaction Level` for each customer cluster.

- **Average Total Spend by Customer Cluster:**
  - **Cluster 0:** Lowest average total spend at $474.

    - **Cluster 1:** Moderate average total spend at $996.
    - **Cluster 2:** Moderate average total spend at $767.
    - **Cluster 3:** Highest average total spend at $1460.
- **Average Satisfaction Level by Customer Cluster (out of 5):**
    - **Cluster 0:** Lowest average satisfaction at 3.33.
    - **Cluster 1:** High average satisfaction at 4.35.
    - **Cluster 2:** High average satisfaction at 4.07.
    - **Cluster 3:** Highest average satisfaction at 4.81.

Analysis: These charts provide clear profiles for each segment:

- **Cluster 3:** High-spending, highly satisfied customers.
- **Cluster 0:** Low-spending, less satisfied customers.
- **Clusters 1 & 2:** Represent intermediate groups, with Cluster 1 showing higher spend and satisfaction than Cluster 2. This reinforces the strong link between spending behavior and satisfaction.

7. Distance from Customers to Cluster Centers

This heatmap (provided twice) shows the distance of three specific customers to the center of each cluster.

- **Customer 10:** Closest to **Cluster 3** (distance 1.70).
- **Customer 42:** Closest to **Cluster 1** (distance 2.74).
- **Customer 87:** Closest to **Cluster 3** (distance 1.94).

Analysis: This visualization is useful for understanding how individual customers are assigned to the most appropriate segment based on their proximity to the cluster centroids.

Overall Conclusion from Visualizations:

The provided images offer a comprehensive view of your customer data. You have a relatively balanced distribution of satisfaction levels. Key features like `Average Rating`, `Total Spend`, and `Age` are strong indicators of satisfaction. Your clustering has successfully identified four distinct customer segments with clear differences in spending habits and satisfaction levels. These insights are valuable for targeted strategies to improve customer experience and drive business outcomes.